```
lab6 : directory management

1.create directory
2.list directory
3.delete directory
4.exit
enter your choice:
1
Enter directory name to create:
harshith
Directory 'harshith' created successfully.

1.create directory
2.list directory
3.delete directory
4.exit
enter your choice:
2
Directories in current location:
harsh/  harshi/  harshith/  harshitha/

1.create directory
2.list directory
3.delete directory
4.exit
enter your choice:
3
enter directory name to delete:
harshi
directory '' deleted successfully.

1.create directory
2.list directory
3.delete directory
4.exit
enter your choice:
4
exiting program...
harshitha@DESKTOP-7ADDEN8:~/first$
```

GNU nano 7.2 labos6.sh

```
#!/bin/bash
echo "lab6 : directory management"
while true
do

echo ""
echo "1.create directory"
echo "2.list directory"
echo "3.delete directory"
echo "4.exit"

echo "enter your choice:"
read choice

case $choice in

1)
   echo "Enter directory name to create:"
   read dirname
   mkdir "$dirname"
   if [ $?-eq 0 ]; then
        echo "Directory '$dirname' created successfully."
   else
        echo "Error creating directory."
   fi
   ;;

2)
   echo "Directories in current location:"
   ls  -d */
   ;;

3)
   echoi "enter directory name to delete:"
   read dirname
```

```
2)
   echo "Directories in current location:"
   ls  -d */
   ;;

3)
   echoi "enter directory name to delete:"
   read dirname
   rmdir"$dirname"
   if [ $? -eq 0 ];then
        echo "directory '$dirnmae' deleted successfully."
   else
        echo "error deleting directory (may not be empty)."
   fi
   ;;

4)
    echo "exiting program..."
    break
    ;;

*)
   echo "invalid choice .try again!"
   ;;

   esac
done
```

GNU nano 7.2 labos7.sh

```
#!/bin/bash

echo "lab 7:memory usage monitor"
echo ""
echo "current memory status (using free command):"
free -h

echo ""
echo "detailed memory info (using top command):"
top -b -n 1 | grep "MiB Mem"

echo ""
echo "Available memory:"
free -h | grep "Mem" | awk '{print "available: "$7}'
```

```
harshitha@DESKTOP-7ADDEN8:~/first$ nano labos6.sh
harshitha@DESKTOP-7ADDEN8:~/first$ nano labos7.sh
harshitha@DESKTOP-7ADDEN8:~/first$ chmod +x labos7.sh
harshitha@DESKTOP-7ADDEN8:~/first$ ./labos7.sh
lab 7:memory usage monitor

current memory status (using free command):
              total        used        free      shared  buff/cache   avail
able
Mem:          3.7Gi       395Mi       3.2Gi       3.5Mi       166Mi       3
.3Gi
Swap:         1.0Gi          0B       1.0Gi

detailed memory info (using top command):
MiB Mem :   3788.2 total,   3297.3 free,    397.3 used,    167.4 buff/cache


Available memory:
available: 3.3Gi
```

GNU nano 7.2 labos8.sh

```bash
#!/bin/bash

echo "lab8: environment variable demo"

#setting variable
name="harshitha"
course="operating systems"

echo ""
echo "Local  variables:"
echo "Name : $name"
echo "course: $course"

#exporting variables
export name
export course

echo ""
echo "using printenv command:"
printenv name
printenv course

echo ""
echo "All environment variables (using env):"
env
```

```
harshitha@DESKTOP-7ADDEN8:~/first$ nano labos8.sh
harshitha@DESKTOP-7ADDEN8:~/first$ chmod +x labos8.sh
harshitha@DESKTOP-7ADDEN8:~/first$ ./labos8.sh
lab8: environment variable demo

Local  variables:
Name : harshitha
course: operating systems

using printenv command:
harshitha
operating systems

All environment variables (using env):
SHELL=/bin/bash
WSL2_GUI_APPS_ENABLED=1
WSL_DISTRO_NAME=Ubuntu
WT_SESSION=2bdced20-aefb-464c-a4fa-56c4a4af9b44
NAME=DESKTOP-7ADDEN8
PWD=/home/harshitha/first
LOGNAME=harshitha
HOME=/home/harshitha
LANG=C.UTF-8
WSL_INTEROP=/run/WSL/300_interop
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;
01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:
st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;
31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31
:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=
01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;3
1:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.
jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace
=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;
31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.avif=01;35:*.jpg=01;35:*.jpeg=01;3
5:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;3
5:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:
*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.
mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.
ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv
```

```
DISPLAY=:0
SHLVL=2
XDG_RUNTIME_DIR=/run/user/1000/
WSLENV=WT_SESSION:WT_PROFILE_ID:
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/game
s:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files (x86)/Common Files
/Oracle/Java/java8path:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/
javapath:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbe
m:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0/:/mnt/c/WINDOWS/System32/O
penSSH/:/mnt/c/Program Files (x86)/NVIDIA Corporation/PhysX/Common:/mnt/c/P
rogram Files/dotnet/:/mnt/c/Program Files/Git/cmd:/mnt/c/Users/Dell/AppData
/Local/Programs/Python/Python314/Scripts/:/mnt/c/Users/Dell/AppData/Local/P
rograms/Python/Python314/:/mnt/c/Users/Dell/AppData/Local/Programs/Python/P
ython313/Scripts/:/mnt/c/Users/Dell/AppData/Local/Programs/Python/Python313
/:/mnt/c/Users/Dell/AppData/Local/Programs/Python/Launcher/:/mnt/c/Users/De
ll/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/Dell/harshi/Microsoft V
S Code/bin:/snap/bin
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
HOSTTYPE=x86_64
course=operating systems
PULSE_SERVER=unix:/mnt/wslg/PulseServer
WT_PROFILE_ID={105648a9-5b68-5b48-b19e-55f5c90bac54}
OLDPWD=/home/harshitha
name=harshitha
_=/usr/bin/env
harshitha@DESKTOP-7ADDEN8:~/first$
```

GNU nano 7.2 lab09.sh

```bash
#!/bin/bash

echo "lab 9: I/O redirection demo"

#output redirection (overwrite)
echo "Hello , this is OS lab">file1.txt
echo "content written to file1.txt"

#append output
echo "appending new line">>file1.txt
echo "content written to file1.txt"


#input redirection
echo ""
echo "reading from file1.txt using input redirection:"
cat < file1.txt


#error redirection
echo ""
echo "trying to read non-existing file:"
cat file_not_exist.txt 2> error.txt
echo "error meeage stored in error.txt"

#display error file
echo ""
echo "error file content:"
cat error.txt
```

```
harshitha@DESKTOP-7ADDEN8:~/first$ nano labos9.sh
harshitha@DESKTOP-7ADDEN8:~/first$ chmod +x labos9.sh
harshitha@DESKTOP-7ADDEN8:~/first$ ./labos9.sh
lab 9: I/O redirection demo
content written to file1.txt
content written to file1.txt

reading from file1.txt using input redirection:
Hello , this is OS lab
appending new line

trying to read non-existing file:
error meeage stored in error.txt

error file content:
cat: file_not_exist.txt: No such file or directory
harshitha@DESKTOP-7ADDEN8:~/first$
```